



K.R. MANGALAM UNIVERSITY

THE COMPLETE WORLD OF EDUCATION

DATABASE MANAGEMENT SYSTEM (ENCS205)

**Bachelor of Technology
in
Computer Science and Engineering**

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LAB SHEET -1

Objective:

- To design ER diagrams and convert entities and relationships into relational tables using proper schema notation.

PART A – COLLEGE DATABASE

STUDENT (USN, SName, Address, Phone, Gender)

```
1  CREATE DATABASE DBMS_Sakshi;
2  USE DBMS_Sakshi;
3
4  CREATE TABLE STUDENT (
5      USN VARCHAR(10) PRIMARY KEY,
6      SName VARCHAR(50),
7      Address VARCHAR(100),
8      Phone VARCHAR(15),
9      Gender VARCHAR(10)
10 );
```

SEMSEC (SSID, Sem, Sec)

```
12 CREATE TABLE SEMSEC (
13     SSID VARCHAR(10) PRIMARY KEY,
14     Sem INT,
15     Sec VARCHAR(5)
16 );
17
```

CLASS (USN, SSID)

```

18 * ⊖ CREATE TABLE CLASS(
19     USN VARCHAR(10),
20     SSID VARCHAR(10),
21     PRIMARY KEY(USN, SSID),
22     FOREIGN KEY (USN) REFERENCES STUDENT(USN),
23     FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
24 );
25

```

IAMARKS (USN, Subcode, SSID, Test1, Test2, Test3, FinalIA)

```

33 * ⊖ CREATE TABLE IAMARKS(
34     USN VARCHAR(10),
35     Subcode VARCHAR(10),
36     SSID VARCHAR(10),
37     Test1 INT,
38     Test2 INT,
39     Test3 INT,
40     FinalIA INT,
41     PRIMARY KEY(USN, Subcode, SSID),
42     FOREIGN KEY (USN) REFERENCES STUDENT(USN),
43     FOREIGN KEY (Subcode) REFERENCES SUBJECT(Subcode),
44     FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
45 );

```

SUBJECT (Subcode, Title, Sem, Credits)

```

26 * ⊖ CREATE TABLE SUBJECT(
27     Subcode VARCHAR(10) PRIMARY KEY,
28     Title VARCHAR(50),
29     Sem INT,
30     Credits INT
31 );
32

```

PART B – COMPANY DATABASE

EMPLOYEE (SSN, Name, Address, Sex, Salary, Superssn, Dno)

```
8  • ⊖ CREATE TABLE EMPLOYEE(  
9      SSN VARCHAR(15) PRIMARY KEY,  
10     Name VARCHAR(50),  
11     Address VARCHAR(100),  
12     Sex VARCHAR(10),  
13     Salary DECIMAL(10,2) NOT NULL,  
14     Superssn VARCHAR(15),  
15     Dno INT,  
16     FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno)  
17 );  
18
```

DEPARTMENT (Dno, Dname, Mgrssn, Mgrstartdate)

```
1  • ⊖ CREATE TABLE DEPARTMENT(  
2      Dno INT PRIMARY KEY,  
3      Dname VARCHAR(50),  
4      Mgrssn VARCHAR(15),  
5      Mgrstartdate DATE  
6  );  
7
```

DLOCATION (Dno, Dloc)

```
19 • ⊖ CREATE TABLE DLOCATION(  
20     Dno INT,  
21     Dloc VARCHAR(50),  
22     FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno)  
23 );  
24
```

PROJECT (PNo, PName, PLocation, DNo)

```

25  CREATE TABLE PROJECT(
26      PNo INT PRIMARY KEY,
27      PName VARCHAR(50),
28      PLocation VARCHAR(50),
29      DNo INT,
30      FOREIGN KEY (DNo) REFERENCES DEPARTMENT(Dno)
31  );
32

```

WORKS_ON (SSN, PNo, Hours)

```

33  CREATE TABLE WORKS_ON(
34      SSN VARCHAR(15),
35      PNo INT,
36      Hours INT,
37      PRIMARY KEY (SSN, PNo),
38      FOREIGN KEY (SSN) REFERENCES EMPLOYEE(SSN),
39      FOREIGN KEY (PNo) REFERENCES PROJECT(PNo)
40  );

```

PART C – CUSTOMER SALE SCHEMA

- Database Schema for a customer-sale scenario
- Customer(Cust id : integer, cust_name: string)

```

1  CREATE TABLE CUSTOMER(
2      Cust_id INT PRIMARY KEY,
3      Cust_name VARCHAR(50)
4  );
5

```

- Item(item_id: integer, item_name:string, price: integer)

```

6  CREATE TABLE ITEM(
7      Item_id INT PRIMARY KEY,
8      Item_name VARCHAR(50),
9      Price INT NOT NULL
10 );

```

- Sale(bill_no:integer,bill_data:date,cust_id: integer, item_id: integer, qty_sold: integer)

```

12  CREATE TABLE SALE(
13      Bill_no INT PRIMARY KEY,
14      Bill_date DATE,
15      Cust_id INT,
16      Item_id INT,
17      Qty_sold INT,
18      FOREIGN KEY (Cust_id) REFERENCES CUSTOMER(Cust_id),
19      FOREIGN KEY (Item_id) REFERENCES ITEM(Item_id)
20  );

```

DDL Questions (Data Definition Language):

- Write SQL statements to create all the above relations with appropriate PRIMARY KEY and FOREIGN KEY constraints.
- Add a new column Email to the STUDENT relation.
- Apply NOT NULL constraints to Salary and Price attributes.
- Rename attribute SName to StudentName in STUDENT relation.
- Drop the DLOCATION relation.

```

1  ALTER TABLE STUDENT ADD Email VARCHAR(50);
2  ALTER TABLE STUDENT RENAME COLUMN SName TO StudentName;
3  DROP TABLE DLOCATION;

```

DML Questions (Data Manipulation Language):

- Insert at least 10 tuples into CUSTOMER, ITEM, and SALE relations.

```

1  INSERT INTO CUSTOMER VALUES
2  (1, 'Aman'), (2, 'Riya'), (3, 'Rahul'), (4, 'Neha'),
3  (5, 'Karan'), (6, 'Pooja'), (7, 'Arjun'),
4  (8, 'Simran'), (9, 'Vikas'), (10, 'Anjali');

```

```

6  INSERT INTO ITEM VALUES
7  (101, 'Pen', 20), (102, 'Book', 150),
8  (103, 'Bag', 500), (104, 'Shoes', 1200),
9  (105, 'Bottle', 250), (106, 'Laptop', 45000),
10 (107, 'Mouse', 300), (108, 'Keyboard', 700),
11 (109, 'Watch', 2000), (110, 'Headphone', 1500);
12

```

```

13  *  INSERT INTO SALE VALUES
14      (1,CURDATE(),1,103,2),
15      (2,CURDATE(),5,104,1),
16      (3,CURDATE(),3,102,3),
17      (4,CURDATE(),5,105,2),
18      (5,CURDATE(),2,110,1),
19      (6,CURDATE(),4,101,5),
20      (7,CURDATE(),6,108,1),
21      (8,CURDATE(),7,106,1),
22      (9,CURDATE(),8,109,1),
23      (10,CURDATE(),9,107,2);

```

List all bills of the current date with customer names and item names.

```

2  *  SELECT s.Bill_no, c.Cust_name, i.Item_name
3      FROM SALE s
4      JOIN CUSTOMER c ON s.Cust_id = c.Cust_id
5      JOIN ITEM i ON s.Item_id = i.Item_id
6      WHERE s.Bill_date = CURDATE();
7

```

- Display total bill amount using quantity and price.

```

9  *  SELECT s.Bill_no, (s.Qty_sold * i.Price) AS Total_Amount
10     FROM SALE s
11     JOIN ITEM i ON s.Item_id = i.Item_id;
12

```

- List customers who purchased products priced above 200.

```

14  *  SELECT DISTINCT c.Cust_name
15     FROM SALE s
16     JOIN CUSTOMER c ON s.Cust_id=c.Cust_id
17     JOIN ITEM i ON s.Item_id=i.Item_id
18     WHERE i.Price > 200;

```

- Count the number of products purchased by each customer.

```
21 * SELECT c.Cust_name, COUNT(*) AS Total_Products
22     FROM SALE s
23     JOIN CUSTOMER c ON s.Cust_id=c.Cust_id
24     GROUP BY c.Cust_name;
```

- List products purchased by customer having Cust_id = 5.

```
27 * SELECT i.Item_name
28     FROM SALE s
29     JOIN ITEM i ON s.Item_id=i.Item_id
30     WHERE s.Cust_id=5;
```

- Create a VIEW showing bill_no, bill_date, cust_id, item_id, price, qty,sold, amount.

```
1 * CREATE VIEW Bill_View AS
2     SELECT s.Bill_no, s.Bill_date, s.Cust_id, s.Item_id,
3           i.Price, s.Qty_sold,
4           (i.Price * s.Qty_sold) AS Amount
5     FROM SALE s
6     JOIN ITEM i ON s.Item_id=i.Item_id;
```

- Create a VIEW showing daily sales date-wise for the last one week.

```
8 * CREATE VIEW Weekly_Sales AS
9     SELECT Bill_date, SUM(Qty_sold * i.Price) AS Daily_Sales
10    FROM SALE s
11    JOIN ITEM i ON s.Item_id=i.Item_id
12    WHERE Bill_date >= CURDATE() - INTERVAL 7 DAY
13    GROUP BY Bill_date;
```